

Reviewer Pack — Minimal Tropical Cyclotomic Polynomial Degree and Resolution...

Entry 632a66835f7d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Tropical Cyclotomic Polynomial Degree and Resolution Proof Width Inequality

Entry ID: 632a66835f7d **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-07 04:25:02 UTC

1. Conjecture

Field A (mathematical branch): Tropical Cyclotomy **Field B** (complexity object): Boolean Satisfiability (Resolution Proof Complexity)

Statement:

For every Boolean satisfiability instance φ with n variables, the degree of the minimal non-trivial tropical cyclotomic polynomial associated with φ is at least $\Omega(n^{1/3})$. Equivalently: for any tropical cyclotomic polynomial $T(\varphi)$ associated with φ , $|T(\varphi)| = \Theta(n^{1/3})$.

Rationale (proposer's reasoning):

The use of tropical cyclotomy, a field less frequently applied to complexity theory, could provide a new perspective on the structure of resolution proof trees. Cyclotomic polynomials have been studied in number theory and algebraic geometry, suggesting they might expose a novel invariant that correlates with resolution proof width.

Taxonomy category: cyclotomy (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): f85fa1248500cac1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all $n \leq 40$ and after generating 30 random Boolean satisfiability instances φ with n variables using 30 different seeds, the degree of the minimal non-trivial tropical cyclotomic polynomial $|T(\varphi)|$ meets or exceeds $\Omega(n^{1/3})$ with a support fraction ≥ 0.95 . The conjecture is falsified if any seed generates an instance where $|T(\varphi)| < n^{1/3}$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "tropical cyclotomy" AND "Boolean satisfiability" AND resolution proof complexity" - "minimal tropical cyclotomic polynomial" AND degree AND $\Omega(n^{(1/3)})$ " - "resolution proof width inequality" AND tropical cyclotomic polynomials

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_instance(n):
        return [random.choice([0, 1]) for _ in range(n)]

    def tropical_cyclotomic_polynomial(instance):
        n = len(instance)
        poly = [Fraction(1)] + [Fraction(0)] * (n - 1)

        for i in range(1, n):
            new_poly = [Fraction(0)] * (n + i - 1)
            for j in range(n):
                if j + i - 1 >= len(new_poly):
                    continue
                new_poly[j + i - 1] += poly[j] * instance[j]
            poly = new_poly

        degree = max(i for i, coeff in enumerate(poly) if coeff != Fraction(0))
        return degree

    n_values = [5, 10, 15, 20, 30, 40]
    total_metric_value = 0
    instances_tested = 0
    n_max = 0
    conjecture_holds = True
    counterexample = ""

    for n in n_values:
```

```

    for _ in range(5): # Ensure at least 30 instances per seed
        instance = generate_boolean_instance(n)
        degree = tropical_cyclotomic_polynomial(instance)
        total_metric_value += degree
        instances_tested += 1
        n_max = max(n_max, n)

        if degree < n ** (1/3):
            conjecture_holds = False
            counterexample = f"n={n}, instance={instance}, degree={degree}"

mean_metric_value = total_metric_value / instances_tested
support_fraction = Fraction(instances_tested - sum(1 for _ in range(instances_tested) if not c

return {
    "metric_name": "Tropical Cyclotomic Polynomial Degree",
    "metric_value": mean_metric_value,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41,

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")

    mean_metric_value = sum(trial["metric_value"] for trial in trials) / len(trials)
    support_fraction = Fraction(sum(1 for trial in trials if trial["conjecture_holds"]), len(trials))

    if all(trial["conjecture_holds"] for trial in trials):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    elif sum(1 for trial in trials if not trial["conjecture_holds"]) / len(trials) >= 0.2:
        print(f"RESULT: FALSIFIED counterexample=\"{trials[0]['counterexample']}\" first_failing_seed={trials[0]['seed']}")
    else:
        print(f"RESULT: INCONCLUSIVE reason=insufficient_evidence n_tested={len(trials)}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_816aafaa.py", line 74, in <module>
    trial_result = run_trial(seed)
                  ~~~~~~

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_816aafaa.py", line 49, in run_trial
    degree = tropical_cyclotomic_polynomial(instance)
            ~~~~~~

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_816aafaa.py", line 36, in tropical_cycl

```

